Type IV

Integrals of the form $\int\sqrt{a^2\pm x^2}\,dx, \int\sqrt{x^2-a^2}\,dx$

Result 11.3

(1) $\int\sqrt{a^2-x^2}\,dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+c$

(2) $\int\sqrt{x^2-a^2}\,dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+c$

(3) $\int\sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+c$

Proof :

(1) Let $I=\int\sqrt{a^2-x^2}\,dx$

Take $u=\sqrt{a^2-x^2}$ then $du=\frac{-2x}{2\sqrt{a^2-x^2}}dx$

$dv=dx\Rightarrow v=x$

Applying Integration by parts, we get

$$\int udv = uv-\int vdu$$

$$\Rightarrow I = x\sqrt{a^2-x^2}-\int\frac{-x^2}{\sqrt{a^2-x^2}}dx$$

$$= x\sqrt{a^2-x^2}-\int\frac{a^2-x^2-a^2}{\sqrt{a^2-x^2}}dx$$

$$= x\sqrt{a^2-x^2}-\int\left(\frac{a^2-x^2}{\sqrt{a^2-x^2}}+\frac{(-a^2)}{\sqrt{a^2-x^2}}\right)dx$$

$$= x\sqrt{a^2-x^2}-\int\sqrt{a^2-x^2}\,dx+\int\frac{a^2}{\sqrt{a^2-x^2}}dx$$

$$= x\sqrt{a^2-x^2}-I+a^2\int\frac{1}{\sqrt{a^2-x^2}}dx$$

$$2I = x\sqrt{a^2-x^2}+a^2\sin^{-1}\left(\frac{x}{a}\right)$$

Therefore, $I=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+c$

Similarly we can prove other two results.

Note 11.3

The above problems can also be solved by substituting $x=a\sin\theta$

Examples 11.41

Evaluate the following :

(i) $\int\sqrt{4-x^2}\,dx$ (ii) $\int\sqrt{25x^2-9}\,dx$ (iii) $\int\sqrt{x^2+x+1}\,dx$ (iv) $\int\sqrt{(x-3)(5-x)}\,dx$

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